Code No: RT31055 **R13** SET - 1

III B. Tech I Semester Supplementary Examinations, October/November- 2020

OPERATING SYSTEMS

(Common to Computer Science and Engineering, Information Technology)

Time: 3 hours Max. Marks: 70

Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B)**
2. Answering the question in **Part-A** is compulsory
3. Answer any **THREE** Questions from **Part-B**

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PART –A (22 Marks)

1. a) Explain any three main functions of an Operating System. [3M]
   b) With a neat diagram, explain various states of a process. [4M]
   c) What are the advantages of inter-process communication and also explain various implementations of inter-process communication? [4M]
   d) Explain various ways to handle a page fault. [4M]
   e) What are the various ways of aborting a process in order to eliminate deadlocks? [4M]
   f) Discuss various issues involved in selecting appropriate disk scheduling algorithm. [3M]

PART –B (48 Marks)

2. a) Discuss the services provided by the operating system for efficient system operation. [8M]
   b) What is a System call? Explain in detail the system call sequence to copy the contents of one file to another file. [8M]

3. a) What are the advantages of inter-process communication? How communication takes place in a shared-memory environment? Explain. [6M]
   b) Assume the following workload in a system: [10M]

Process	Arrival Time	Burst Time
P1	5	5
P2	4	6
P3	3	7
P4	1	9
P5	2	2
P6	6	3

Draw a Gantt chart illustrating the execution of these jobs using Round robin scheduling algorithm and also Calculate the average waiting time and average turnaround time.

4. a) Define the hardware instruction Test And Set() and Swap() also give the mutual exclusion implementation with both. [6M]
   b) How semaphores can be used to deal with n-process critical section problem? Explain. [10M]

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5. a) Given free memory partitions of 100 K, 500 K, 200 K, 300 K, and 600 K (in order), how would each of the First-fit, Best-fit, and Worst-fit algorithms place processes of 212 K, 417 K, 112 K, and 426 K (in order)? [8M]

   b) Define Virtual Memory. Explain the process of converting virtual addresses to physical addresses with a neat diagram. [8M]

6. a) How does deadlock avoidance differ from deadlock prevention? Write about deadlock avoidance algorithm in detail. [8M]

   b) Explain the bankers algorithm with a suitable example. [8M]

7. A disk drive has 200 cylinders, numbered 0 to 199. The drive is currently serving a request at cylinder 53. The queue of pending requests, in FIFO order, is [16M]

   98, 183, 37, 122, 14, 124, 65, 67.

   Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms?

   i) FCFS ii) SSTF iii) SCAN iv) LOOK v) C-SCAN vi) C-LOOK.

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